Exercice 1. on considère l'équation de la chaleur sur $\Omega =]0,1[$

$$\frac{\partial u}{\partial t}(t,x) - \frac{\partial^2 u}{\partial x^2}(t,x) = 0, \quad t>0, \ x \in \Omega$$

$$u(t,0) = u(t,1) = 0 \quad \text{(Dirichlet)}$$

$$u(0,x) = u_0(x), \ x \in \Omega$$

a) soit $v \in \mathcal{C}^1([0,1])$ tq $v(0) = 0$

Mq v vérifie l'inégalité de Poincaré : $\|v\|_2 \leq c \cdot \|v'\|_2$ ($\exists c$ tq ...)

On a $|v(x)| = |v(0) + \int_0^x 1 \cdot v'(y)\,dy|$

$\leq \int_0^1 1 \cdot |v'(y)|\,dy$

$\leq \|1\|_2 \cdot \|v'\|_2$

$\Rightarrow \|v\|_\infty \leq \|v'\|_2$

$\Rightarrow \int_0^1 |v(x)|^2 dx \leq \|v\|_\infty^2 \cdot \int_0^1 1\,dx$

$\Rightarrow \|v\|_2^2 \leq \|v'\|_2^2$

$\Rightarrow \|v\|_2 \leq 1 \cdot \|v'\|_2$

b) On définit l'énergie : $E(t) = \int_0^1 u^2(t,x)\,dx$. Mq $\frac{1}{2}E'(t) = -\int_0^1 \left(\frac{\partial u}{\partial x}(t,x)\right)^2 dx \leq 0$ ($\Rightarrow$ énergie monotone ↘)

Or, u sol $\Rightarrow \int_0^1 \left(\underbrace{\frac{\partial u}{\partial t} \cdot u}_{\frac{\partial u}{\partial t}(t,x)\cdot u(t,x) = \frac{1}{2}\frac{\partial}{\partial t}(u^2(t,x))} - \underbrace{\frac{\partial^2 u}{\partial x^2} \cdot u}_{\text{IPP}}\right) dx = 0$

$\Rightarrow \frac{1}{2}\int_0^1 \frac{\partial}{\partial t}(u^2(t,x))dx - \left[u \frac{\partial u}{\partial x}\right]_0^1 + \int_0^1 \left(\frac{\partial u}{\partial x}\right)^2 dx = 0$

cf CL de Dirichlet $\Rightarrow u(t,0) = u(t,1) = 0$ ($\forall t>0$)

Si $\exists g \in L^1(X, \beta, \mu)$ tq $\left|\frac{\partial f}{\partial t}(t,x)\right| \leq g(x)$, pp $x \in X$ et $t \in]t_0 - \varepsilon, t_0 + \varepsilon[$ $\forall \varepsilon > 0$

Alors F est dérivable en t_0 et $F'(t_0) = \int_X \frac{\partial f}{\partial t}(t_0, x)\,d\mu(x)$

c) Mq $E(t) \leq E(0) \cdot e^{-2t}$, $t \geq 0$.

On a $\frac{1}{2}E'(t) = -\int_0^1 \left(\frac{\partial u}{\partial x}\right)^2 dx \leq -\int_0^1 u^2 dx = -E(t)$

$\Rightarrow E'(t) \leq -2E(t)$

Supposons $E(t) > 0$ (on sait déjà que $E(t) \geq 0$) :

$\frac{E'(t)}{E(t)} \leq -2 = \underbrace{\int_0^t \frac{E'(s)}{E(s)} ds}_{= [\ln|E(s)|]_0^t = \ln \frac{E(t)}{E(0)} > 0} \leq -2t$

$\Rightarrow E(t) \leq e^{-2t}$ (cf exp ↗)

Rq : si $u_0 \neq 0$ (ie on n'a pas $u_0(x) = 0, x \in \Omega$)

on mq $\int_0^1 u^2(t,x)\,dx > 0$

Si $u_0 \equiv 0$, $E(0) = 0 \Rightarrow E(t) = 0$ $\forall t > 0 \Rightarrow E(t) = 0 \leq 0 = E(0)e^{-2t}$ $t \geq 0$

cf ↘ et $E(t) \geq 0$

Exercice 2

a) On considère l'équation des ondes en dim 1 ($\Omega \subset \mathbb{R}$)

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} - \dfrac{\partial^2 u}{\partial x_2} = 0 \ , \ t>0 \quad x \in \Omega \\ u(t,x) = 0 \ , \ x \in \partial\Omega \ , \ t>0 \\ u(0,x) = u_0(x) \ , \ x \in \Omega \\ \dfrac{\partial u}{\partial t}(0,x) = u_1(x) \ , \ x \in \Omega \end{cases}$$

On suppose $\Omega =]0,1[$, mq l'énergie

$E(t) := \int_0^1 [(\frac{\partial u}{\partial t})^2 + (\frac{\partial u}{\partial x})^2]\,dx$ est cst

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial t}\right) \overset{\text{Schwarz}}{=} \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x}\right)$$

Montrons pour cela que $E'(t) = 0$ si u sol.

Or u sol $\Rightarrow$ $\displaystyle\int_0^1 \Big(\underbrace{\frac{\partial^2 u}{\partial t^2}\frac{\partial u}{\partial t}}_{\frac{1}{2}\frac{\partial}{\partial t}\left[\left(\frac{\partial u}{\partial t}\right)^2\right]} - \frac{\partial^2 u}{\partial x^2}\cdot\frac{\partial u}{\partial t}\Big)dx = 0$

$$\int_0^1 \frac{\partial^2 u}{\partial x^2}\cdot\frac{\partial u}{\partial t}\,dx = \Big[\frac{\partial u}{\partial x}\cdot\underbrace{\cancel{\frac{\partial u}{\partial t}}}_{0\,(*)}\Big]_0^1 - \int_0^1 \underbrace{\frac{\partial u}{\partial x}\cdot\frac{\partial^2 u}{\partial x \partial t}}_{\frac{1}{2}\frac{\partial}{\partial t}\left[\left(\frac{\partial u}{\partial x}\right)^2\right]}\,dx$$

$$\Rightarrow 0 = \underbrace{\frac{1}{2}\int_0^1 \frac{\partial}{\partial t}\left[\left(\frac{\partial u}{\partial t}\right)^2 + \left(\frac{\partial u}{\partial x}\right)^2\right]dx}_{=\frac{1}{2}E'(t)}$$

(*) on a bien $\frac{\partial u}{\partial t}(t,0) = \frac{\partial u}{\partial t}(t,1) = 0$

Puisque Dirichlet $\Rightarrow \forall t>0 \ \begin{cases} u(t,0)=0 \\ u(t,1)=0 \end{cases}$

$\Rightarrow \forall t>0 \ \begin{cases} \frac{\partial u}{\partial t}(t,0)=0 \\ \frac{\partial u}{\partial t}(t,1)=0 \end{cases}$

b) En déduire l'unicité de solution. Soient v et w deux sol de notre équation, posons $u := v - w$

par linéarité de l'EDP, u vérifie $\dfrac{\partial^2 u}{\partial t^2} - \dfrac{\partial^2 u}{\partial x^2} = 0$ et $\begin{cases} u(t,x) = 0 \text{ si } x \in \partial\Omega = \{0,1\} \\ u(0,x) = 0 \\ \frac{\partial u}{\partial t}(0,x) = 0 \end{cases}$

D'après ce qui précède, l'énergie associée à u est conservée et $E(t) = cst = E(0) = \int_0^1 [(\frac{\partial u}{\partial t}(0,x))^2 + (\frac{\partial u}{\partial x}(0,x))^2]dx$

De plus, $(\forall x \in \Omega)$: $u(0,x) = 0$

$\Rightarrow (\forall x \in \Omega) : \frac{\partial u}{\partial x}(0,x) = 0$

$\Rightarrow (\forall t \geq 0) : E(t) = 0 = \int_0^1 [\underbrace{(\frac{\partial u}{\partial t}(t,x))^2}_{\geq 0} + \underbrace{(\frac{\partial u}{\partial x}(t,x))^2}_{\geq 0}]dx$

$\Rightarrow (\forall (t,x) \in \mathbb{R}_+^* \times \Omega) : \frac{\partial u}{\partial t}(t,x) = \frac{\partial u}{\partial x}(t,x) = 0$

$\Rightarrow (\exists c \in \mathbb{R}) : u(t,x) = c$ Dirichlet $\Rightarrow u(t,x) = u(t,0) = u(t,1) = 0 \ t \geq 0, x \in \Omega$

D'où l'unicité

c) Sur $\Omega = \mathbb{R}$ vérifier que $u(t,x) := \frac{1}{2}(u_0(x+t) + u_0(x-t)) + \frac{1}{2}(U_1(x+t) - U_1(x-t))$

où U_1 est une primitive de u_1 (cf $\frac{\partial u}{\partial t}(0,x) = u_1(x)$)

est solution de

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} - \dfrac{\partial^2 u}{\partial x^2} = 0 \ , \ t>0 \ , \ x \in \Omega = \mathbb{R} \\ u(0,x) = u_0(x) \ , \ x \in \Omega = \mathbb{R} \\ \dfrac{\partial u}{\partial t}(0,x) = u_1(x) \ , \ x \in \Omega = \mathbb{R} \end{cases}$$

NB : $\partial\Omega = \partial\mathbb{R} = \emptyset$: Pas de CL

$$\frac{\partial u}{\partial t}(t,x) = \frac{1}{2}\left(u_0'(x+t) - u_0'(x-t)\right) + \frac{1}{2}\left(u_1(x+t) + u_1(x-t)\right)$$

$$\frac{\partial^2 u}{\partial t^2}(t,x) = \frac{1}{2}\left(u_0''(x+t) + u_0''(x-t)\right) + \frac{1}{2}\left(u_1'(x+t) - u_1'(x-t)\right)$$

$$\frac{\partial u}{\partial x}(t,x) = \frac{1}{2}\left(u_0'(x+t) + u_0'(x+t)\right) + \frac{1}{2}\left(u_1(x+t) - u_1(x-t)\right)$$

$$\frac{\partial^2 u}{\partial x^2}(t,x) = \frac{1}{2}\left(u_0''(x+t) + u_0''(x-t)\right) + \frac{1}{2}\left(u_1'(x+t) - u_1'(x-t)\right)$$

$$\Rightarrow \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0$$

De plus, $u(0,x) = u_0(x)$, $\frac{\partial u}{\partial t}(0,x) = u_1(x)$

Exercice 3

$$\begin{cases} i\dfrac{\partial u}{\partial t}(t,x) + \dfrac{\partial^2 u}{\partial x^2}(t,x) - v(x)\,u(t,x) = 0 \\ u(t,x) \xrightarrow[|x|\to\infty]{} 0 \ , \ t \geq 0 \\ \dfrac{\partial u}{\partial x}(t,x) \xrightarrow[|x|\to\infty]{} 0 \\ u(0,x) = u_0(x) \ , \ x \in \Omega := \mathbb{R} \end{cases}$$

a) Soit $v : \mathbb{R}^n \to \mathbb{C}$ dérivable, mq :

$$\frac{1}{2}\frac{\partial |v|^2}{\partial y_i} = \mathcal{R}e\left(\frac{\partial v}{\partial y_i}\bar{v}\right)$$

on a pour tout $y \in \mathbb{R}^n$, $|v(y)|^2 = v(y)\cdot\bar{v}(y)$

$\Rightarrow$ $|v|^2$ est dérivable qd v l'est, produit et composée d'applications dérivables.

(cf $|v(y)|^2 = v_1^2(y) + v_2^2(y) = (v_1(y) + i v_2(y))\cdot\overbrace{(v_1(y) - i v_2(y))}^{\bar{v}(y)}$ en notant $v(y) = v_1(y) + i v_2(y) \simeq \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}(y)$)

De plus, le produit (dans $\mathbb{C}$) étant bilinéaire, on a

$$\frac{\partial}{\partial y_i}(v(y)\cdot\bar{v}(y))$$

$$= \frac{\partial v}{\partial y_i}(y)\cdot\bar{v}(y) + \underbrace{v(y)\cdot\overbrace{\frac{\partial(\bar{v})}{\partial y_i}(y)}^{=\overline{\frac{\partial v}{\partial y_i}(y)}}}_{\overline{\frac{\partial v}{\partial y_i}(y)\cdot\bar{v}(y)}}$$

$$= 2\mathcal{R}e\left(\frac{\partial v}{\partial y_i}(y)\cdot\bar{v}(y)\right) \in \mathbb{R}$$

b) Soit u sol de l'EDP de Schrodinger, mq on a conservation de l'énergie

$$\int_{\mathbb{R}} |u(t,x)|^2\,dx = \int_{\mathbb{R}} |u_0(x)|^2\,dx$$

Or si u solution, on a : $\displaystyle\int_{\mathbb{R}} \left(i\frac{\partial u}{\partial t}(t,x) + \frac{\partial^2 u}{\partial x^2}(t,x) - \overbrace{v(x)}^{\in\mathbb{R}}\,u(t,x)\right)\cdot\bar{u}(t,x)\,dx = 0$

donc $\displaystyle i\int_{\mathbb{R}} \frac{\partial u}{\partial t}\bar{u}\,dx = -\int_{\mathbb{R}} \frac{\partial^2 u}{\partial x^2}\bar{u}\,dx + \underbrace{\int_{\mathbb{R}} v|u|^2\,dx}_{\in\mathbb{R}}$

$$-\int_{\mathbb{R}} \frac{\partial^2 u}{\partial x^2}\bar{u}\,dx = -\left[\cancel{\frac{\partial u}{\partial x}\bar{u}}^{\,0}\right]_{-\infty}^{+\infty} + \int_{\mathbb{R}} \frac{\partial \bar{u}}{\partial x}\frac{\partial u}{\partial x}\,dx = \int_{\mathbb{R}} \underbrace{\left|\frac{\partial u}{\partial x}\right|^2}_{\geq 0}\,dx \in \mathbb{R}$$

Donc $\displaystyle i\int_{\mathbb{R}} \frac{\partial u}{\partial t}\bar{u}\,dx \in \mathbb{R}$ Donc $\displaystyle\int_{\mathbb{R}} \frac{\partial u}{\partial t}\bar{u}\,dx \in i\mathbb{R}$ Donc $\displaystyle\mathcal{R}e\left(\int_{\mathbb{R}} \frac{\partial u}{\partial t}\bar{u}\,dx\right) = 0$

$$= \int_{\mathbb{R}} \mathcal{R}e\left(\frac{\partial u}{\partial t}\bar{u}\right)dx = \frac{1}{2}\frac{\partial |u|^2}{\partial t}$$

Donc $\displaystyle\int_{\mathbb{R}} \frac{1}{2}\frac{\partial}{\partial t}(|u(t,x)|^2)\,dx = 0$

Sous les bonnes hypothèses, $\displaystyle\frac{1}{2}\frac{d}{dt}\int_{\mathbb{R}} |u(t,x)|^2\,dx = 0$

Donc $\displaystyle\frac{1}{2}\int_{\mathbb{R}} |u(t,x)|^2 = \text{cst} \underset{\text{c.i}}{=} \frac{1}{2}\int_{\mathbb{R}} |u_0(x)|^2\,dx$

c) Montrer finalement que $\int \left(\left| \frac{\partial u}{\partial x}(t,x) \right|^2 + v(x) \cdot |u(t,x)|^2 \right) dx = \int \left(\left| \frac{\partial u_0}{\partial x}(x) \right|^2 + v(x) \cdot |u_0(x)|^2 \right) dx$

On a si u sol : $\int_{\mathbb{R}} \left(i \frac{\partial u}{\partial t}(t,x) + \frac{\partial^2 u}{\partial x^2}(t,x) - \overset{\in \mathbb{R}}{v(x)} \cdot u(t,x) \right) \times \frac{\partial \bar{u}}{\partial t}(t,x)\, dx = 0$

Donc $i \int_{\mathbb{R}} \frac{\partial u}{\partial t}(t,x) \frac{\partial \bar{u}}{\partial t}(t,x)\, dx + \int_{\mathbb{R}} \left(\frac{\partial^2 u}{\partial x^2}(t,x) - v(x) \cdot u(t,x) \right) \times \frac{\partial \bar{u}}{\partial t}\, dx = 0$

Donc $i \int_{\mathbb{R}} \left| \frac{\partial u}{\partial t}(t,x) \right|^2 dx + \int_{\mathbb{R}} \left(\frac{\partial^2 u}{\partial x^2}(t,x) - v(x) \cdot u(t,x) \right) \times \frac{\partial \bar{u}}{\partial t}\, dx = 0$

Donc $\mathcal{R}e \left(i \int_{\mathbb{R}} \left| \frac{\partial u}{\partial t} \right| dx + \int_{\mathbb{R}} \left(\frac{\partial^2 u}{\partial x^2} - v \cdot u \right) \frac{\partial \bar{u}}{\partial t}\, dx \right) = 0$

De plus $\int_{\mathbb{R}} \frac{\partial^2 u}{\partial x^2} \cdot \frac{\partial \bar{u}}{\partial t}\, dx = \left[\frac{\partial u}{\partial x} \cdot \frac{\partial \bar{u}}{\partial t} \right]_{-\infty}^{+\infty} - \int_{\mathbb{R}} \frac{\partial u}{\partial x} \cdot \frac{\partial^2 \bar{u}}{\partial x \partial t}\, dx$

Donc $\int_{\mathbb{R}} \left[\underbrace{\mathcal{R}e \left(\underbrace{\frac{\partial u}{\partial x} \frac{\partial^2 \bar{u}}{\partial x \partial t}}_{\frac{\partial u}{\partial x} \frac{\partial}{\partial t} \left(\frac{\partial \bar{u}}{\partial x} \right)} \right)}_{= \frac{1}{2} \frac{\partial}{\partial t} \left(\left| \frac{\partial \bar{u}}{\partial x} \right|^2 \right)} + v(x) \underbrace{\mathcal{R}e \left(u \frac{\partial \bar{u}}{\partial t} \right)}_{= \frac{1}{2} \frac{\partial}{\partial t} \left(|u|^2 \right)} \right] dx = 0$

Donc $\int_{\mathbb{R}} \frac{\partial}{\partial t} \left[\left| \frac{\partial u}{\partial x} \right|^2 + v(x) |u|^2 \right] dx = 0$